PHASE 2 · BEGINNER

Day 02

Variables & Data Types

Session length: 2 hours

4	2	20+
Data Types	Core Tools	Practice Problems

Session Overview

Goal: Understand Python's core data types and how to declare, inspect, convert, and use variables to capture and display information.

Outcome: By the end of this session, students can confidently create variables of any built-in type, check a value's type, take user input, and convert between types safely.

Time	Segment
0:00 – 0:15	Recap Day 1 + intro to variables
0:15 – 0:45	int, float, str, bool + type()
0:45 – 1:00	Break
1:00 – 1:30	input() / print() and formatting
1:30 – 1:50	Type casting (int(), str(), float())
1:50 – 2:00	Wrap-up + preview Day 3

■ TIP

Use lots of rupee/Mumbai-flavoured examples today — students relate faster to ■250 chai budgets than abstract numbers.

Prerequisites & Preparation

Requirement	Status
Day 1 environment setup completed	Required
VS Code + Python extension working	Required
Comfortable running a .py file	Required

■■ WATCH OUT

Instructor checklist: open VS Code, create day02_variables.py, have the Python interactive console ready for live demos.

Step-by-Step Instruction Guide

STEP 1 — Variables & Core Data Types

"A variable is a labelled box that stores a value. Python figures out the type automatically — this is called dynamic typing."

```
name = "Priya"
age = 23
price = 149.50
is_member = True
# Output:
Priya 23 149.5 True
```

■ DONE WHEN

Done when students can name the 4 core types: int, float, str, bool.

STEP 2 — type() and Checking Values

```
print(type(name))
print(type(price))
# Output:
<class 'str'>
<class 'float'>
```

STEP 3 — input() and print()

```
city = input("Enter your city: ")
print("You live in", city)
# Terminal:
$ python day02_variables.py
Enter your city: Mumbai
You live in Mumbai
```

Common Error	Fix
NameError: name not defined	Variable not assigned before use
input() always returns str	Wrap with int()/float() if needed

STEP 4 — Type Casting

```
budget_text = "500"
budget = int(budget_text)
total = budget + 100
print("Total budget: Rs.", total)
# Output:
Total budget: Rs. 600
```

■ DONE WHEN

Done when students can convert str to int/float and back without errors.

Session Wrap-Up

Student checklist

- ■ Created variables of all 4 core types
- ■ Used type() to inspect a variable
- ■ Took input from the user and printed formatted output
- ■ Converted between str, int, and float

Key concepts recap

Python variables are dynamically typed; int/float/str/bool are the four foundational types; input() always returns a string; type casting bridges between types.

■ **TIP**

Next session (Day 3): Control Flow — if/elif/else, comparison and logical operators, and the match statement.

Quick Reference Card

Command	Purpose
type(x)	Check the type of a value
int(x) / float(x) / str(x)	Convert between types
input("prompt")	Read text input from user
print(a, b, sep=', ')	Print multiple values